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# Ch.12 Human Skeletal and Muscular Systems

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

## MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

**1. Which structure belongs to the appendicular skeleton?**

**Answer: (d) Humerus bone** — The humerus is a limb bone; ethmoid, ribs and lumbar vertebrae are axial.

**2. What does “muscle fibre/myofibre” refer to?**

**Answer: (b) A cell** — A muscle fibre is a single multinucleated muscle cell.

**3. Which structure extends the entire length of a muscle fibre?**

**Answer: (b) Myofibril** — Myofibrils run the length of the muscle fibre and contain sarcomeres.

**4. Actin (thin) filaments are made of which proteins?**

**Answer: (d) Actin, tropomyosin and troponin** — Thin filaments contain actin plus regulatory tropomyosin and troponin.

**5. Z-lines provide attachment for ends of which filaments?**

**Answer: (a) Actin filaments** — Thin actin filaments anchor at Z-lines.

**6. A sarcomere lies between what?**

**Answer: (c) Two Z-lines** — The sarcomere is the contractile unit from one Z-line to the next.

**7. Which structure releases $Ca^{2+}$ to trigger contraction?**

**Answer: (d) Sarcoplasmic reticulum** — The SR stores and releases calcium ions.

**8. Which statement describes the sliding-filament model?**

**Answer: (b) Myosin pulls on actin so the sarcomere shortens** — Filaments slide past each other; they do not shorten themselves.

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**9. When a muscle fibre shortens, which also shortens?**

**Answer: (c) Sarcomere** — Sarcomeres shorten as Z-lines move closer; actin/myosin lengths remain constant.

**10. Which event correctly describes muscle contraction?**

**Answer: (c) ATP is used to detach the myosin head from actin** — ATP binding detaches myosin; hydrolysis re-cocks the head for the next power stroke.

**11. Tendons connect bone to what?**

**Answer: (c) Muscle** — Tendons attach muscle to bone; ligaments attach bone to bone.

**12. What is true about antagonistic muscle pairs?**

**Answer: (c) They allow opposing movements** — One muscle contracts while the other relaxes, producing opposite actions.

## SHORT QUESTIONS

**1. Name three bone cell types and their functions.**

| Cell | Function |
|---|---|
| Osteoblasts | Build bone matrix and promote mineralization. |
| Osteocytes | Mature bone cells that maintain bone tissue. |
| Osteoclasts | Resorb/break down bone for remodelling and calcium release. |

**2. Name the bones of the cranium.**

Frontal, parietal (2), temporal (2), occipital, sphenoid and ethmoid.

**3. List the bones in the five vertebral groups.**

Cervical (7), thoracic (12), lumbar (5), sacral (5 fused into the sacrum), coccygeal (4 fused into the coccyx) — 33 vertebrae in total.

**4. What bones make the rib cage?**

Sternum, 12 pairs of ribs, and 12 thoracic vertebrae.

**5. Name the bones of the pectoral and pelvic girdles.**

Pectoral: clavicle and scapula. Pelvic: two hip bones, each formed by ilium, ischium and pubis.

**6. Name the bones of the forelimbs and hindlimbs.**

Forelimb: humerus, radius, ulna, carpals, metacarpals, phalanges. Hindlimb: femur, patella, tibia, fibula, tarsals, metatarsals, phalanges.

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### 7. What is a fibrous joint? Give examples.

A fibrous joint is joined directly by dense fibrous connective tissue (mainly collagen) and permits no movement. Examples: sutures between skull bones, the joint between tibia and fibula, and teeth in their jaw sockets.

### 8. Name the steps of bone repair.

> Fracture haematoma → Fibrocartilaginous callus → Bony callus → Remodelling

The chapter describes ossification and remodelling (osteoblasts build bone, osteoclasts resorb it); the standard fracture-repair sequence above answers the exercise prompt.

### 9. What skeletal structures are affected by osteoarthritis?

The articular cartilage at the ends of bones softens and disintegrates, so bone ends rub together. It commonly affects knee, hip and intervertebral joints; joint space narrows and bone spurs may form.

### 10. List the major parts of a skeletal muscle fibre.

Sarcolemma, sarcoplasm, multiple nuclei, myofibrils, sarcomeres, sarcoplasmic reticulum, T-tubules, mitochondria and glycogen granules.

### 11. What are I-band, A-band and H-zone?

I-band: light region containing only thin actin filaments. A-band: dark region containing thick myosin filaments (and overlapping actin). H-zone: central lighter part of A-band with only myosin.

### 12. Describe the antagonistic arrangement of skeletal muscles.

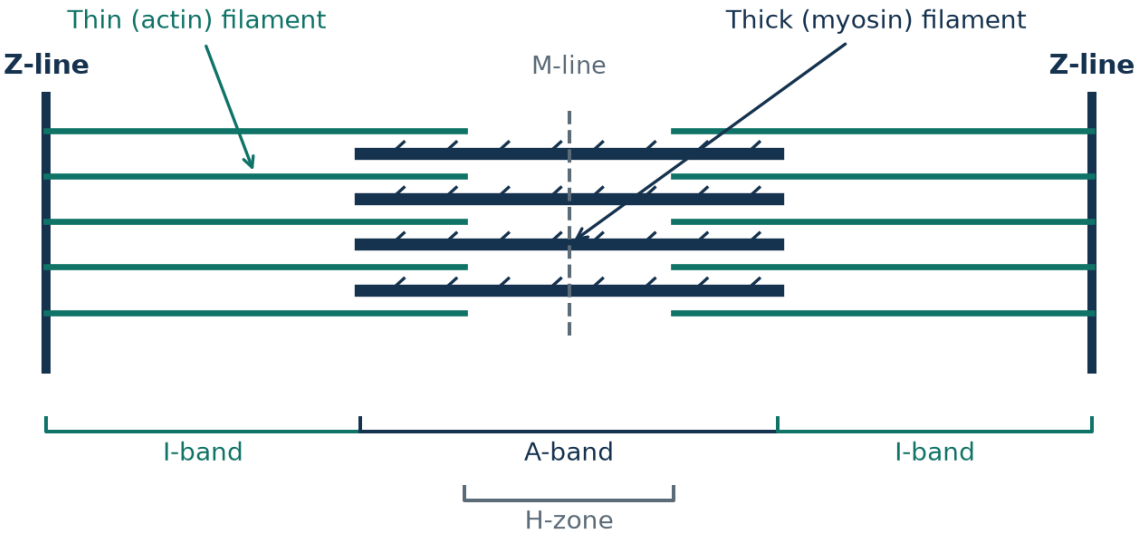
Muscles work in opposing pairs: when one contracts (agonist), the other relaxes (antagonist). Example: biceps flexes the elbow while triceps relaxes; triceps extends while biceps relaxes.

### 13. Justify: ligaments are elastic while tendons are hard.

Ligaments join bone to bone and contain more elastic fibres, allowing slight stretch and joint stability. Tendons join muscle to bone and are dense, tough collagenous cords designed to transmit strong pulling forces with little stretch.

### 14. Draw a labelled sarcomere diagram.

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Labelled sarcomere: Z-lines, actin, myosin, I-band, A-band, H-zone and M-line.

## 15. Differentiate the following pairs.

### i. Compact and spongy bone

| Compact bone | Spongy bone |
|---|---|
| **Structure:** Dense osteons/Haversian systems. | **Structure:** Trabeculae with marrow spaces. |
| **Location:** Outer shaft of bones. | **Location:** Ends of long bones and inside flat bones. |

### ii. Axial and appendicular skeleton

| Axial | Appendicular |
|---|---|
| **Parts:** Skull, vertebral column, rib cage. | **Parts:** Limbs and girdles. |
| **Role:** Support/protect central organs. | **Role:** Movement and manipulation. |

### iii. True, false and floating ribs

| True ribs (1–7) | False ribs (8–10) | Floating ribs (11–12) |
|---|---|---|
| Attach directly to the sternum by separate costal cartilages. | Attach indirectly to the sternum via the common costal cartilage of rib 7. | No anterior attachment to the sternum. |

### iv. Rheumatoid arthritis and osteoarthritis

| Rheumatoid arthritis | Osteoarthritis |
|---|---|
| **Cause:** Autoimmune inflammation of synovium. | **Cause:** Wear-and-tear degeneration of cartilage. |
| **Pattern:** Symmetrical, systemic. | **Pattern:** Often weight-bearing joints, localized. |

### v. Fibrous and cartilaginous joints

| Fibrous | Cartilaginous |
|---|---|
| **Connecting tissue:** Dense fibrous tissue. | **Connecting tissue:** Cartilage. |
| **Movement:** Little/no movement. | **Movement:** Slight movement. |
| **Example:** Skull sutures. | **Example:** Intervertebral discs/pubic symphysis. |

### vi. Cartilaginous and synovial joints

| Cartilaginous | Synovial |
|---|---|
| **Cavity:** No joint cavity. | **Cavity:** Joint cavity with synovial fluid. |
| **Movement:** Limited. | **Movement:** Freely movable. |
| **Example:** Vertebral discs. | **Example:** Knee, shoulder. |

### vii. Osteoblasts and osteocytes

| Osteoblasts | Osteocytes |
|---|---|
| **Role:** Bone-forming cells. | **Role:** Mature bone-maintaining cells. |
| **Location:** Bone surfaces. | **Location:** Lacunae within matrix. |

### viii. Tropomyosin and troponin

| Tropomyosin | Troponin |
|---|---|
| **Role:** Blocks myosin-binding sites on actin at rest. | **Role:** Binds $Ca^{2+}$ and moves tropomyosin. |
| **Structure:** Long filamentous protein. | **Structure:** Complex of three subunits. |

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### ix. Ligament and tendon

| Ligament | Tendon |
|---|---|
| **Connects:** Bone to bone. | **Connects:** Muscle to bone. |
| **Property:** More elastic. | **Property:** Tough, less elastic. |

### x. Tetany and tetanus

| Tetany | Tetanus |
|---|---|
| **Cause:** Electrolyte imbalance, often low $Ca^{2+}$. | **Cause:** Bacterial infection by Clostridium tetani toxin. |
| **Effect:** Involuntary spasms due to overexcitable nerves. | **Effect:** Severe muscle stiffness/spasms, often jaw/neck. |

## LONG QUESTIONS

### 1. Explain the structure of bone.

Bone is a connective tissue reinforced with calcium salts and specialized cells. Its surface is covered by periosteum. Beneath it lies compact bone, whose basic units are Haversian systems (osteons).

- **Lamellae:** concentric layers of mineralized matrix containing collagen and calcium phosphate crystals; collagen gives flexibility, crystals give hardness.
- **Lacunae and osteocytes:** mature bone cells sit in lacunae and connect via canaliculi.
- **Haversian canal:** central canal containing blood vessels, nerves and lymphatics; perpendicular Volkmann's canals connect them.
- **Spongy bone:** latticework of bony spikes (trabeculae) beneath compact bone; light but strong.
- **Bone marrow:** red marrow produces blood cells; yellow marrow stores fat and can convert to red marrow after severe blood loss.

Bone cells: osteoblasts form bone, osteocytes maintain it, and osteoclasts resorb it for remodelling and calcium release.

### 2. Describe the structure of the three types of cartilage.

Cartilage is a flexible connective tissue surrounded by perichondrium. Chondrocytes lie in lacunae within a matrix of collagen, elastin and proteoglycans. It lacks blood vessels, so it heals slowly.

| Hyaline cartilage | Fibrocartilage | Elastic cartilage |
|---|---|---|
| **Matrix:** glassy, fine collagen fibres. | **Matrix:** dense collagen bundles. | **Matrix:** elastic fibres plus collagen. |

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| Hyaline cartilage | Fibrocartilage | Elastic cartilage |
|---|---|---|
| **Properties:** smooth, reduces friction. | **Properties:** strong, resists tension/compression. | **Properties:** flexible, returns to shape. |
| **Location:** nose, trachea, articular surfaces of joints. | **Location:** intervertebral discs, pubic symphysis, menisci. | **Location:** external ear, epiglottis. |

### 3. Write causes and symptoms of joint dislocation, spondylosis and sciatica.

| Condition | Cause | Symptoms |
|---|---|---|
| Dislocation | Bones forced out of normal joint position by trauma/impact; may tear muscles, ligaments and tendons. | Swelling, intense pain, immobility, deformity of the joint. |
| Spondylosis | Degeneration of vertebrae, discs, ligaments or cartilage; aging, genetics, trauma, poor posture, obesity; bone outgrowths press nerves/spinal cord. | Back or neck pain, stiffness, reduced range of motion; common in lumbar and cervical regions. |
| Sciatica | Compression/irritation of the sciatic nerve, often by a herniated/bulging disc; also trauma, infection, inflammation, spondylosis. | Pain from lower back through buttock down leg/foot; tingling, numbness, weakness or difficulty moving leg/foot. |

### 4. Describe types of arthritis with causes, symptoms and treatments.

Arthritis is inflammation of joints, causing pain and stiffness; redness, warmth and swelling may also occur.

| Type | Cause | Symptoms | Treatment |
|---|---|---|---|
| Osteoarthritis | Articular cartilage softens and disintegrates (wear-and-tear); affects knee, hip, intervertebral joints. | Pain, stiffness, reduced movement. | Weight control, exercise, physiotherapy, analgesics; severe cases may need joint replacement. |
| Rheumatoid arthritis | Autoimmune inflammation of synovial membrane; commonly wrist and hands. | Swollen painful joints, fatigue, deformity; often symmetrical. | Anti-inflammatory drugs, DMARDs, physiotherapy. |

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| Type | Cause | Symptoms | Treatment |
|---|---|---|---|
| Gouty arthritis | Uric acid builds up in blood and forms crystals in joints; commonly big toe, also knees/wrists/fingers. | Sudden severe pain, redness, swelling. | Urate-lowering drugs, anti-inflammatories, diet control. |

## 5. Describe the three types of muscles.

| Skeletal muscle | Smooth muscle | Cardiac muscle |
|---|---|---|
| **Appearance:** striated, cylindrical fibres. | **Appearance:** non-striated, spindle-shaped cells. | **Appearance:** striated, branched fibres with intercalated discs. |
| **Nuclei:** many per cell. | **Nuclei:** one per cell. | **Nuclei:** usually one per cell. |
| **Control:** voluntary. | **Control:** involuntary. | **Control:** involuntary. |
| **Location/function:** attached to bones; moves skeleton. | **Location/function:** walls of hollow organs; moves substances. | **Location/function:** heart wall; pumps blood. |

## 6. Explain the ultrastructure of skeletal muscle.

A skeletal muscle is made of bundles of muscle fibres (cells) enclosed by connective tissue; at the ends this tissue forms tendons attaching muscle to bone.

- **Muscle fibre:** cylindrical, multinucleated cell enclosed by sarcolemma; cytoplasm is sarcoplasm.
- **Myofibrils:** 4–20 threadlike structures per fibre, made of thick (myosin) and thin (actin) filaments.
- **Sarcomere:** contractile unit between two Z-lines; thin filaments anchor at Z-lines and overlap thick filaments.
- **Bands:** A-band (dark, thick filaments), I-band (light, thin filaments), H-zone (central A-band with no overlap), M-line stabilizes thick filaments.
- **Membrane systems:** sarcoplasmic reticulum stores $Ca^{2+}$; T-tubules conduct impulses into the fibre.

## 7. Write a detailed note on the sliding-filament model of muscle contraction.

Nerve impulse reaches sarcolemma → T-tubules → SR releases $Ca^{2+}$ → $Ca^{2+}$ binds troponin → tropomyosin moves, exposing actin sites → Myosin heads bind actin (cross-bridges) → Power stroke: heads release ADP + Pi and pull actin toward centre → ATP binds myosin → head detaches → ATP hydrolysis re-cocks head; cycle repeats

Thin filaments slide deeper into A-bands, so the sarcomere shortens: Z-lines move closer, I-bands and H-zone narrow/disappear, while A-bands do not shorten. Actin and myosin filaments themselves keep constant length. Relaxation occurs when $Ca^{2+}$ is pumped back into the SR and tropomyosin re-blocks binding sites.

### 8. Explain antagonistic muscle action in movement of the knee joint.

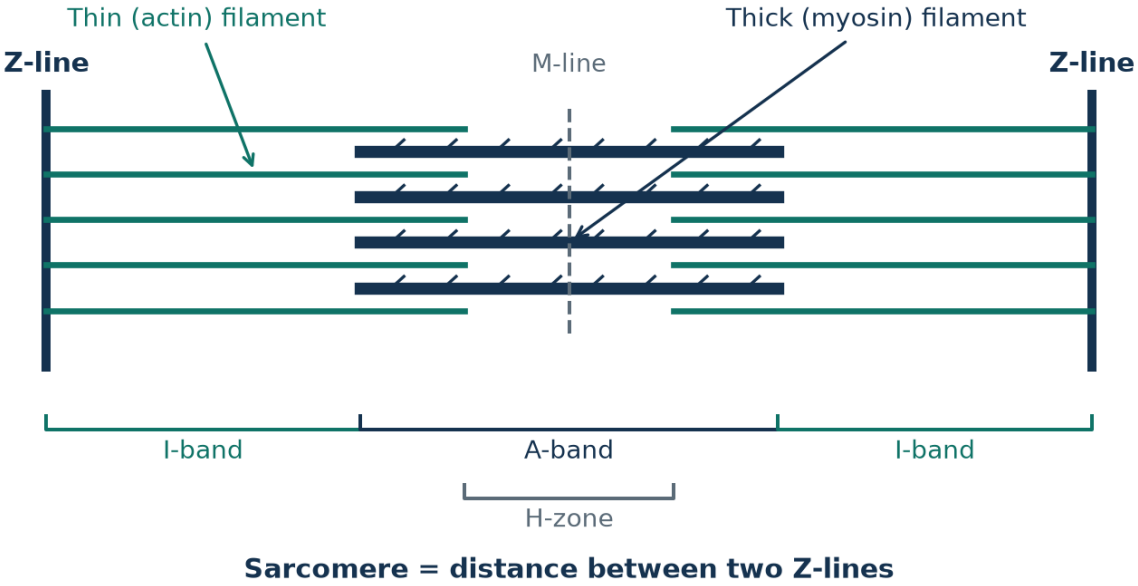
The knee joint lies between the femur and the tibia/fibula. Muscles work in antagonistic pairs to bend and straighten it.

- **Flexion (bending):** hamstrings (three muscles at the back of the thigh) contract; quadriceps relax. Hamstrings originate at the pelvic girdle/top of femur and insert on upper tibia/fibula.
- **Extension (straightening):** quadriceps (four muscles at the front of the thigh) contract; hamstrings relax. Quadriceps originate at ilium/femur and insert via the patella onto the tibia.

Because muscles can only pull, opposing pairs are needed to move the bone in both directions.

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### 9. Draw a labelled sarcomere diagram.




Sarcomere from Z-line to Z-line: thin actin filaments anchor at Z-lines; thick myosin filaments lie centrally; I-band contains only actin, A-band contains myosin plus overlap, H-zone contains only myosin, and the M-line stabilizes the centre.

During contraction the Z-lines move closer, the I-band and H-zone narrow, and the A-band remains constant in length.

### 10. Describe causes and symptoms of muscle fatigue, cramps and tetany.

| Condition | Cause | Symptoms |
|---|---|---|
| Muscle fatigue | Prolonged/intense activity or disease; ATP depletion, metabolic wastes (lactate, $H^+$, reactive oxygen), impaired $Ca^{2+}$ handling, micro-damage to fibres. | Pain, decreased strength, reduced endurance; improves with rest. |
| Muscle cramp | Sudden involuntary contraction; dehydration, salt imbalance, overuse/injury, certain drugs (diuretics), diabetes/liver disease/nerve damage. | Painful muscle knot, usually legs/feet; lasts seconds to minutes; relieved by stretching/massage, heat/cold. |
| Tetany | Hyperexcitable nerves due to hypocalcaemia (vitamin D deficiency, renal failure, thyroid disorders) or low magnesium. | Involuntary spasms of hands, feet, face or larynx; treated by correcting electrolyte imbalance. |

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### 11. Justify how the skeleton acts as a system of rods and levers.

Bones act as rigid rods/levers that give structure, support and protection. Joints serve as fulcrums (pivot points), muscles provide the effort (force), and the weight or resistance being moved is the load.

For example, when lifting a weight: the elbow joint is the fulcrum, the biceps provides the effort, the forearm bones act as the lever, and the weight in the hand is the load. This arrangement allows movement and mechanical advantage, so the skeleton functions as a system of rods and levers.

### 12. Justify why muscles pull but do not push.

Muscle fibres generate force only by contracting and shortening. When a muscle contracts, it pulls on tendons and thus moves bones toward the muscle. It cannot actively lengthen to push a bone away.

Return movement is produced by an antagonistic muscle pulling in the opposite direction, or by gravity/elastic recoil. Therefore muscles work in pairs: one pulls to flex, the other pulls to extend. This is why muscles pull but do not push.

## INQUISITIVE QUESTIONS

### 1. Why is calcium essential for bone structure and muscle contraction?

Calcium salts harden bone matrix. In muscle, $Ca^{2+}$ binds troponin to expose actin binding sites, enabling cross-bridge cycling. Thus calcium is needed for both support and contraction.

### 2. Why does the skeleton have rigid bones and flexible joints instead of one solid structure?

Rigid bones provide support and protection, while joints allow movement. A single solid structure would protect but prevent locomotion, manipulation and flexibility.

### 3. Why do muscles work in antagonistic pairs rather than alone?

Muscles can only pull by contracting. An opposing pair is needed to return a bone to its original position and produce controlled movement in both directions.

### 4. Why do prolonged inactivity or space travel cause muscle atrophy and bone weakening?

Without mechanical loading, muscle protein synthesis falls and muscles waste. Bones lose mineral density because osteoclast activity exceeds bone formation under reduced gravitational/mechanical stress.

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